

Report 3: Fixed-Wing Position Control

Part I: Complete Original System

Part II: Every Soaring-Mode Modification and Its Impact

PX4 Autosozaring Implementation — Defence Technical Report

Radhouene — Fixed-Wing Autosozaring Project

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List of Symbols

A superscript $(\cdot)^*$ denotes a *setpoint* (commanded value). A dot $(\dot{\cdot})$ denotes the time derivative. Bold symbols ($\mathbf{v}$) denote 2-D or 3-D vectors. A hat $(\hat{\cdot})$ (in guidance context) denotes a unit vector.

Position and Velocity

Symbol	Meaning	Unit
$\mathbf{p} = (x, y)$	Aircraft ground position in local NED North-East plane	m
$\mathbf{v}_g = (v_N, v_E)$	Aircraft ground velocity (North, East)	m/s
$\mathbf{w} = (w_N, w_E)$	Wind velocity vector (North, East)	m/s
$\mathbf{v}_a = \mathbf{v}_g - \mathbf{w}$	Aircraft airspeed vector (ground velocity minus wind)	m/s
$V_a = \ \mathbf{v}_a\ $	Airspeed magnitude	m/s
$v_{g,x}$	Body-frame forward ground speed (from estimator)	m/s
$V_{g,\min}$	Minimum forward ground speed (FW_GND_SPD_MIN)	m/s
h	Altitude AMSL, computed as $h = -z_{\text{NED}} + h_{\text{ref}}$	m
h_{ref}	EKF reference altitude (local_pos.ref_alt)	m

Attitude Angles

Symbol	Meaning	Unit
ϕ	Bank (roll) angle, positive right-wing-down	rad
ϕ^*	Roll setpoint	rad
$\phi_{\max}$	Maximum roll limit (FW_R_LIM)	rad
θ	Pitch angle, positive nose-up	rad
θ^*	Pitch setpoint (from TECS)	rad
ψ	Yaw (heading) angle, measured clockwise from North	rad
$\mathbf{q}$	Unit quaternion representing current attitude	—
$\mathbf{q}^*$	Attitude setpoint quaternion published to inner-loop	—
R	Direction Cosine Matrix (DCM) from body to NED	—

Airspeed

Symbol	Meaning	Unit
V_{EAS}	Equivalent airspeed: corrected for air density	m/s
V_{TAS}	True airspeed: actual speed relative to air mass	m/s
$k = k_{\text{EAS2TAS}} = \sqrt{\rho_0/\rho}$	EAS-to-TAS conversion factor, ≥ 1	—
V_{EAS}^*	EAS setpoint after <code>adapt_airspeed_setpoint()</code>	m/s
V_{trim}	Trim (cruise) EAS from performance model	m/s
$V_{\text{EAS},\min}, V_{\text{EAS},\max}$	EAS flight envelope bounds from performance model	m/s
$V_{\text{TAS},\min}, V_{\text{TAS},\max}$	Corresponding TAS bounds $= k \cdot V_{\text{EAS},\min/\max}$	m/s
k_w	Wind-based airspeed scaling coefficient (FW_WIND_ARSP_SC)	—

NPFG Guidance

Symbol	Meaning	Unit
$\hat{\mathbf{t}}$	Unit tangent vector along the reference path	—
$\hat{\mathbf{n}} = \hat{\mathbf{t}}^\perp$	Unit normal to the path (points left of direction of travel)	—
e_y	Signed cross-track error: positive = aircraft is right of path	m
η	Track-error angle between airspeed vector and desired heading	rad
L	NPFG look-ahead distance (scales with airspeed and period)	m
T_{NPFG}	NPFG guidance period parameter (NPFG_PERIOD)	s
ζ	NPFG damping ratio (NPFG_DAMPING)	—
a_{lat}	Lateral (horizontal) acceleration command from NPFG	m/s ²
R	Loiter circle radius (positive = clockwise)	m
$f_{\text{run}} \in [0, 1]$	NPFG can-run confidence factor (1 = full confidence)	—

TECS Interface (as seen from this module)

Symbol	Meaning	Unit
h^*	Altitude setpoint passed to TECS	m
θ_{TECS}^*	Pitch setpoint returned by <code>get_tecs_pitch()</code>	rad
$\delta_{t,\text{TECS}}^*$	Throttle returned by <code>get_tecs_thrust()</code>	—
$n_z = 1/\cos \phi$	Normal load factor injected into TECS for stall floor	—

adapt_airspeed_setpoint Notation

Symbol	Meaning	Unit
$V_{\text{EAS},\text{in}}^*$	Input EAS setpoint (from waypoint or mission)	m/s
$V_{\text{EAS},\text{out}}^*$	Slew-rate limited EAS setpoint passed to TECS	m/s
$V_{\text{EAS},\text{min},\text{wind}}^*$	Wind-adjusted minimum EAS = $V_{\text{EAS},\text{min}} + k_w \ \mathbf{w}\ $	m/s
r_V	Airspeed setpoint slew rate (ASPD_SP_SLEW_RATE, default 1 m/s ²)	m/s ²

Manual Control

Symbol	Meaning	Unit
δ_{pitch}	Pitch stick input, $\in [-1, +1]$	—
δ_{throttle}	Throttle stick input, $\in [-1, +1]$	—
δ_{roll}	Roll stick input, $\in [-1, +1]$	—
δ_{yaw}	Yaw stick input, $\in [-1, +1]$	—
h_{stick}^*	Height rate setpoint from pitch stick	m/s
V_{stick}^*	Airspeed setpoint from throttle stick	m/s
ψ_{hold}	Latched heading in heading-hold mode	rad
$\mathbf{p}_{\text{hold}}$	Latched NE position in heading-hold mode	m

GPS Failsafe

Symbol	Meaning	Unit
T_{gpsf}	Fixed-bank loiter duration before descend (NAV_GPSF_LT, default 30 s)	s
ϕ_{gpsf}	Fixed open-loop bank angle (NAV_GPSF_R, default 15°)	deg
h_{descent}	Hard-coded descent rate in GPS failsafe = -0.5 m/s	m/s

Control Mode Flags (set by set_control_mode_current)

Symbol / Name	Condition
FW_POSCTRL_MODE_AUTO	Normal auto: position + position setpoint valid
FW_POSCTRL_MODE_AUTO_TAKEOFF	Takeoff setpoint type
FW_POSCTRL_MODE_AUTO_LANDING_STRAIGHT	Land type + valid previous WP
FW_POSCTRL_MODE_AUTO_LANDING_CIRCULAR	Land type, no previous WP
FW_POSCTRL_MODE_AUTO_PATH	Offboard with velocity setpoint
FW_POSCTRL_MODE_AUTO_ALTITUDE	GPS failsafe phase 1 (fixed-bank loiter)
FW_POSCTRL_MODE_AUTO_CLIMBRATE	GPS failsafe phase 2 (descend)
FW_POSCTRL_MODE_MANUAL_ALTITUDE	Manual + altitude hold
FW_POSCTRL_MODE_MANUAL_POSITION	Manual + heading hold

Soaring-Mode Symbols (Part II)

Symbol	Meaning	Unit
ϕ_{cmd}	Commanded bank angle for thermal (bank_angle_cmd)	deg
$ \phi_{\text{cmd}} $	Magnitude of commanded bank angle	deg
R_{min}	Minimum safe loiter radius at speed V and bank ϕ	m
R_{eff}	Effective loiter radius: $\max(R_{\text{companion}}, R_{\text{min}})$	m
R_c	Companion-provided loiter radius (loiter_radius_m)	m
T_{ramp}	Throttle ramp duration (FW_GLIDE_RAMP_T, default 3 s)	s
t_s	Entry ramp start timestamp (_soaring_entry_ramp_start_us)	μs
t_e	Exit ramp start timestamp (_soaring_ramp_start_us)	μs
$\delta_{t,\text{pre}}$	TECS throttle just before glide entry (_soaring_pre_glide_thrust)	—
$s_{\text{exit}}(t)$	Exit ramp scale factor = $\min((t - t_e)/T_{\text{ramp}}, 1)$	—
$h_{\text{min}}, h_{\text{max}}$	Soaring altitude floor/ceiling (FW_ALT_MIN, FW_ALT_MAX)	m
h_{hyst}	Altitude ceiling hysteresis band (FW_ALT_HYST)	m
$V_{\text{EAS},\text{glide}}^*$	Glide EAS setpoint injected into TECS for soaring modes	m/s
a, b	Glide polar coefficients: sink rate $w_s = aV^2 + b$	—
$V_{\text{polar}}^* = \sqrt{b/a}$	Best-glide EAS from polar (Mode 2)	m/s
$\mathbf{q}^* = \text{Euler}(\phi^*, \theta^*, \psi)$	Mode 4 attitude setpoint quaternion	—

Mathematical Operators

Operator	Meaning	Notes
<code>clamp(x, a, b)</code>	Saturate: $\max(a, \min(b, x))$	Also <code>constrain()</code> in PX4
<code>lerp(a, b, t)</code>	Linear interpolation: $(1 - t)a + tb$	$t \in [0, 1]$
<code>Euler(ϕ, θ, ψ)</code>	ZYX Euler angles to unit quaternion	Roll, pitch, yaw order
<code>atan2(y, x)</code>	Four-quadrant arctangent	Returns angle in $(-\pi, \pi]$
<code>sign(x)</code>	Signum: +1 if $x > 0$, else -1	
<code>$\ \mathbf{v}\$</code>	Euclidean norm of vector $\mathbf{v}$	
<code>ϵ</code>	Small positive guard value against division by zero	
<code>max(a, b), min(a, b)</code>	Standard maximum / minimum	
<code>x, <code>fabsf(x)</code></code>	Absolute value	

Part I

Original Fixed-Wing Position Control System

1 Module Identity and Scheduling

File: `src/modules/fw_pos_control/FixedwingPositionControl.cpp`

Task trigger: The module is scheduled by the `local_position` uORB topic; `Run()` is called every time a new position estimate is published (nominally 50 Hz, $\Delta t = 20$ ms).

Interval guard:

$$\Delta t_{\text{eff}} = \text{clamp}\left(\frac{t_k - t_{k-1}}{10^6}, \Delta t_{\text{min}}, \Delta t_{\text{max}}\right) \quad [0.01 \text{ s}, 0.1 \text{ s}] \quad (1)$$

The clamp prevents integration blow-up during long-pauses (start-up, debugger break) and avoids division by zero.

Output topics written each cycle (if not idle):

- `vehicle_attitude_setpoint` ($\mathbf{q}^*, \delta_t$)
- `tecs_status` (debug / logging)
- `position_controller_status`
- `autosoaring_status` (FMU $\rightarrow$ companion: `soar:off` / `inhibit` / `watchdog`)
- `landing_gear` (on state change)
- `flight_phase_estimation`

2 Run() — Main Execution Loop

`Run()` is the entry point called by the scheduler. Its complete execution order each cycle:

1. **Exit check:** if `should_exit()`, unregister callback and clean up.
2. **Parameter update:** if `parameter_update` uORB updated, call `parameters_update()`.
3. **DDS soaring poll:** check `autosoaring_control` uORB; apply `inhibit/override/watchdog` logic; optionally republish `autosoaring_status` at 1 Hz while the `post-soar:off` DDS `inhibit` is active (see Section 26).
4. **Global position:** update (ϕ, λ) from `vehicle_global_position`.
5. **Altitude reference:** $h = -z_{\text{NED}} + h_{\text{ref}}$ where h_{ref} is the EKF reference altitude.
6. **EKF reset handling** (manual modes only):
 - Altitude reset: call `_tecs.handle_alt_step()` to prevent a TECS altitude-error spike.

- XY reset: clear heading-hold flag.

7. Setpoint ingestion:

- Offboard (`flag_control_offboard_enabled`): convert `trajectory_setpoint` NED position/velocity into `position_setpoint_triplet` in lat/lon/alt. If velocity + acceleration are valid, compute signed loiter radius from lateral acceleration:

$$R = -\frac{v_x^2 + v_y^2}{\|a_\perp\|} \cdot \text{sign}(\hat{\mathbf{v}}_H \times \hat{\mathbf{a}}_\perp) \quad (2)$$

- Normal: subscribe `position_setpoint_triplet`; set validity flags for pre-v/current/next. Reset altitude FOH counter `_min_current_sp_distance_xy`.

8. Sensor polls (in order): `airspeed_poll()`, `manual_control_setpoint_poll()`, `vehicle_attitude_poll()`, `vehicle_command_poll()`, `vehicle_control_mode_poll()`, `wind_poll()`.

9. Air data: update air density ρ from `vehicle_air_data`; push density-dependent TECS parameters:

$$\dot{h}_{\text{climb,max}} = f(\rho), \quad \dot{h}_{\text{sink,min}} = f(\rho) \quad (3)$$

10. Land detection: update `_landed`.

11. Vehicle status: update; clear VTOL backtransition state if not in transition.

12. Ground speed vector: (v_N, v_E) from local position.

13. Mode selection: `set_control_mode_current()`.

14. In-air states: `update_in_air_states()`.

15. TECS speed weight restore: write `FW_T_SPDWEIGHT` back to TECS (may have been overridden in landing).

16. NPFG period restore: write `NPFG_PERIOD` back (may have been overridden in takeoff).

17. Default outputs: `flaps = 0`, `spoilers = 0`, `gear = keep`, `fw_control_yaw_wheel = false`, `reset_integral = false`.

18. Landing/takeoff state reset (if not in respective mode).

19. Mode dispatch switch (17 cases — see Section 6).

20. Post-dispatch: throttle slew-rate gate; `status_publish()`; `landing_status_publish()`; gear publication; actuator setpoint publish; TECS debug publish.

3 Sensor Poll Functions

3.1 airspeed_poll()

1. If `FW_USE_AIRSPD` enabled and `airspeed_validated` has a new message:

- Check both `calibrated_airspeed_m_s` and `true_airspeed_m_s` are finite.
- Set `_airspeed_eas` (EAS) and compute EAS-to-TAS factor:

$$k = \text{clamp}\left(\frac{V_{\text{TAS,meas}}}{V_{\text{EAS,meas}}}, 0.9, 2.0\right) \quad (4)$$

- The clamp prevents out-of-range ISA atmosphere states.
- Record `_time_airspeed_last_valid`.

2. If no update for > 1 s: mark airspeed invalid.

3. On validity change: call `_tecs.enable_airspeed(valid)` to switch TECS between airspeed and no-airspeed control paths.

3.2 wind_poll()

1. If `wind` uORB updated: store (w_N, w_E) ; set valid if both components are finite.

2. Timeout: if last update $> \text{WIND_EST_TIMEOUT}$, invalidate.

3. If wind invalid: zero (w_N, w_E) to prevent stale wind from contaminating NPFG (which uses wind for track-keeping).

3.3 manual_control_setpoint_poll()

1. Copy latest stick input from `manual_control_setpoint`.

2. Normal mapping:

$$\text{height_rate_stick} = \delta_{\text{pitch}}, \quad \text{airspeed_stick} = \delta_{\text{throttle}} \quad (5)$$

3. Alternate mapping (`FW_POS_STK_CONF` bit 0 set):

$$\text{height_rate_stick} = -\delta_{\text{throttle}}, \quad \text{airspeed_stick} = \delta_{\text{pitch}} \quad (6)$$

4. Timeout: if last update > 1 s, zero both to prevent stale input integrating into height or airspeed.

3.4 vehicle_attitude_poll()

1. Read quaternion $\mathbf{q}$ from `vehicle_attitude`; read body rates from `vehicle_angular_velocity`.

2. Tailsitter correction: if VTOL tailsitter, rotate DCM by 90° pitch offset before extracting Euler angles.

3. Extract (ϕ, θ, ψ) using `Eulerf(Dcmf(q))`.

4. Compute body-frame acceleration: $\mathbf{a}_b = R^\top(a_N, a_E, a_D)^\top$. Store $\mathbf{a}_{b,x}$ for ground-speed undershoot detection.
5. Compute body-frame velocity: store $\mathbf{v}_{b,x}$.
6. Update TECS load factor:

$$n_z = \frac{1}{\cos \phi}, \quad \texttt{_tecs.set_load_factor}(n_z) \quad (7)$$

This feeds the banked-stall speed floor (see Report 2, M9).

3.5 vehicle_control_mode_poll()

1. If updated: check if arming state just changed (was `flag_armed = false`, now true).
2. On arming: call `reset_takeoff_state()` and `reset_landing_state()` to flush stale integrator states.

3.6 vehicle_command_poll()

This function processes vehicle commands from uORB in a while-loop (multiple commands may arrive between cycles).

DO_GO_AROUND Aborts landing if in `FW_POSCTRL_MODE_AUTO` with `SETPOINT_TYPE_LAND` and position control is enabled. Calls `updateLandingAbortStatus(ABORTED_BY_OPERATOR)`, which sets a flag that makes `control_auto_loiter()` fly level until above a clearance altitude buffer.

DO_CHANGE_SPEED If `param1 = SPEED_TYPE_AIRSPPEED` and `param2 > 0`:

- Auto mode: overrides `pos_sp_triplet.current.cruising_speed` (used by `adapt_airspeed_setpoint`).
- Manual mode: stores `_commanded_manual_airspeed_setpoint` (used by `get_manual_airspeed_setpoint`).

DO_SET_MODE In the original system: no specific handling in `FixedwingPositionControl` (handled by Commander).

4 parameters_update()

Called on every `parameter_update` event. Pushes all parameter values into submodule instances.

4.1 NPFG Parameters

PX4 Parameter	NPFG Method
NPFG_PERIOD	<code>setPeriod()</code>
NPFG_DAMPING	<code>setDamping()</code>
NPFG_EN_PERIOD_LB/UB	<code>enablePeriodLB/UB()</code>
NPFG_EN_MIN_GSP	<code>enableMinGroundSpeed()</code>
NPFG_EN_TRACK_KEEPING	<code>enableTrackKeeping()</code>
NPFG_EN_WIND_REG	<code>enableWindExcessRegulation()</code>
FW_GND_SPD_MIN	<code>setMinGroundSpeed()</code>
NPFG_ROLL_TC	<code>setRollTimeConst()</code>
FW_R_LIM	<code>setRollLimit()</code> (converted to rad)
NPFG_SW_DST_MULT	<code>setSwitchDistanceMultiplier()</code>
NPFG_PERIOD_SF	<code>setPeriodSafetyFactor()</code>

4.2 TECS Parameters

PX4 Parameter	TECS Method
FW_T_CLMB_MAX (via perf model, ρ)	<code>set_max_climb_rate()</code>
FW_T_SINK_MAX	<code>set_max_sink_rate()</code>
min sink rate (via perf model, ρ)	<code>set_min_sink_rate()</code>
FW_T_SPDWEIGHT	<code>set_speed_weight()</code>
trim/min/max EAS (via perf model)	<code>set_equivalent_airspeed_*</code>
FW_T_THR_DAMPING	<code>set_throttle_damp()</code>
FW_T_THR_INTEG	<code>set_integrator_gain_throttle()</code>
FW_T_I_GAIN_PIT	<code>set_integrator_gain_pitch()</code>
FW_THR_SLEW_MAX	<code>set_throttle_slewrates()</code>
FW_T_VERT_ACC	<code>set_vertical_accel_limit()</code>
FW_T_RLL2THR	<code>set_roll_throttle_compensation()</code>
FW_T_PTCH_DAMP	<code>set_pitch_damping()</code>
FW_T_H_ERROR_TC	<code>set_altitude_error_time_constant()</code>
FW_T_FAST_ALT_ERR	<code>set_fast_descend_altitude_error()</code>
FW_T_HRATE_FF	<code>set_altitude_rate_ff()</code>
FW_T_TAS_ERROR_TC	<code>set_airspeed_error_time_constant()</code>
STE_RATE_TC	<code>set_ste_rate_time_const()</code>
SEB_RATE_FF	<code>set_seb_rate_ff_gain()</code>
ASPD_FMT_STD_DEV	<code>set_airspeed_measurement_std_dev()</code>
ASPD_RATE_STD_DEV	<code>set_airspeed_rate_measurement_std_dev()</code>
ASPD_PROC_NOISE	<code>set_airspeed_filter_process_std_dev()</code>

The performance model (`FixedwingPerformanceModel`) computes density-dependent limits from the aerodynamic polars and `FW_T_CLMB_MAX`, `FW_T_SINK_MIN`. On every `parameters_update()` call, `_performance_model.runSanityChecks()` validates that `min < trim < max` airspeed.

5 set_control_mode_current(): Mode Decision Tree

This function runs every cycle *after* all polls, selecting one of eight control modes:

Mode	Condition
FW_POSCTRL_MODE_OTHER	Rotary-wing vehicle type without transition, or no matching condition.
FW_POSCTRL_MODE_AUTO_PATH	Offboard enabled + position enabled + finite NED velocity setpoint.
FW_POSCTRL_MODE_AUTO	Auto + position enabled + valid current setpoint (or IDLE type).
FW_POSCTRL_MODE_AUTO_TAKEOFF	As above, but setpoint type is TAKEOFF (fixed-wing only).
FW_POSCTRL_MODE_AUTO_LANDING_STRAIGHT	As above, type LAND + prev setpoint valid.
FW_POSCTRL_MODE_AUTO_LANDING_CIRCULAR	As above, type LAND + prev not valid.
FW_POSCTRL_MODE_AUTO_ALTITUDE	Auto + climb-rate enabled + armed + position <i>not</i> valid; GPS failsafe phase 1 (fixed-bank loiter for NAV_GPSF_LT seconds).
FW_POSCTRL_MODE_AUTO_CLIMBRATE	Same conditions, after NAV_GPSF_LT timeout; GPS failsafe phase 2 (emergency descent).
FW_POSCTRL_MODE_MANUAL_POSITION	Manual + position enabled.
FW_POSCTRL_MODE_MANUAL_ALTITUDE	Manual + altitude enabled.
FW_POSCTRL_MODE_TRANSITION_TO_HOVER_*	VTOL back-transition.

VTOL takeoff bypass: If VTOL and in transition while setpoint type is TAKEOFF, the type is rewritten to POSITION and mode is set to FW_POSCTRL_MODE_AUTO to use the standard waypoint follower instead of the launch sequence.

Arming guard: Failsafe altitude/climbrate modes only activate when the vehicle is armed (`flag_armed = true`) to prevent triggering during pre-arm checks.

6 Run() Mode Dispatch Switch

After `set_control_mode_current()`, a switch dispatches to the appropriate control function:

```

1 switch (_control_mode_current) {
2   case FW_POSCTRL_MODE_AUTO:
3     control_auto(...); break;
4   case FW_POSCTRL_MODE_AUTO_ALTITUDE:
5     control_auto_fixed_bank_alt_hold(...); break;
6   case FW_POSCTRL_MODE_AUTO_CLIMBRATE:
7     control_auto_descend(...); break;
8   case FW_POSCTRL_MODE_AUTO_LANDING_STRAIGHT:
9     control_auto_landing_straight(...); break;
10  case FW_POSCTRL_MODE_AUTO_LANDING_CIRCULAR:
11    control_auto_landing_circular(...); break;
12  case FW_POSCTRL_MODE_AUTO_PATH:
13    control_auto_path(...); break;
14  case FW_POSCTRL_MODE_AUTO_TAKEOFF:
15    control_auto_takeoff(...); break;
16  case FW_POSCTRL_MODE_MANUAL_POSITION:
17    control_manual_position(...); break;
18  case FW_POSCTRL_MODE_MANUAL_ALTITUDE:

```

```

19   control_manual_altitude(...); break;
20   case FW_POSCTRL_MODE_TRANSITION_TO_HOVER_LINE_FOLLOW:
21     control_backtransition_line_follow(...); break;
22   case FW_POSCTRL_MODE_TRANSITION_TO_HOVER_HEADING_HOLD:
23     control_backtransition_heading_hold(); break;
24 }

```

7 NPFG: Nonlinear Path Following Guidance

7.1 Core Guidance Law

NPFG computes a lateral acceleration command based on the error between the current airspeed vector and the desired track direction.

Track-error angle η : For a line segment $A \rightarrow B$ with unit tangent $\hat{\mathbf{t}}$:

$$e_y = \hat{\mathbf{n}} \cdot (\mathbf{p} - \mathbf{A}) \quad (\text{signed cross-track error}) \quad (8)$$

$$\eta = \text{atan2}(\mathbf{v}_a, \hat{\mathbf{t}}) + \arcsin\left(\text{clamp}\left(\frac{-e_y}{L}, -1, 1\right)\right) \quad (9)$$

where L is the look-ahead distance.

Look-ahead length:

$$L = \frac{T_{\text{NPFG}} V_a}{2\pi/\zeta} = f(\text{NPFG period, damping}) \quad (10)$$

Bounded below by a minimum ground speed term and above by a maximum.

Lateral acceleration command:

$$a_{\text{lat}} = \frac{2V_a^2}{L} \sin \eta \quad (11)$$

For loiter circle: The centripetal acceleration is added:

$$a_{\text{lat}} += \text{sign}(d) \cdot \frac{V_a^2}{R} \quad (12)$$

where $d = \pm 1$ for CW/CCW.

Roll setpoint:

$$\phi^* = \arctan\left(\frac{a_{\text{lat}}}{g}\right) \quad (13)$$

Clamped to $[-\phi_{\text{max}}, \phi_{\text{max}}]$ from FW_R_LIM.

7.2 getCorrectedNpfgRollSetpoint()

NPFG reports a “can-run factor” $f_{\text{run}} \in [0, 1]$ that measures confidence based on:

- Velocity estimate validity (`xy_valid` in local position).
- Wind estimate validity (`_wind_valid`).

The corrected roll:

$$\phi_{\text{corrected}}^* = f_{\text{run}} \cdot \phi_{\text{NPFG}}^* \quad (14)$$

If $f_{\text{run}} < 0.9$ for more than `ROLL_WARNING_TIMEOUT` (2s), a user warning event is emitted. This prevents the aircraft from making large rolls in GPS-degraded conditions.

Roll slew rate: After NPFG, the roll setpoint is filtered through a slew-rate limiter (`FW_PN_R_SLEW_MAX`, default 90 deg/s) to prevent step demands.

8 adapt_airspeed_setpoint()

Called before every TECS invocation. Six processing steps:

1. **Wind-based minimum airspeed scaling:** If airspeed valid, wind valid, and `FW_WIND_ARSP_SC > 0`:

$$V_{\text{EAS},\text{min}} \leftarrow \min(V_{\text{EAS},\text{min}} + k_w \|\mathbf{w}\|, V_{\text{EAS},\text{max}}) \quad (15)$$

This increases the minimum airspeed in strong winds to maintain climb performance and avoid stall.

2. **NaN/zero setpoint fallback:** If the mission waypoint cruising speed is not finite or ≤ 0 : $V_{\text{EAS}}^* \leftarrow V_{\text{trim}}$.
3. **Ground-speed undershoot correction:** If wind is *not* valid (cannot estimate it):

$$\text{if } v_{g,x} < V_{g,\text{min}} : \quad V_{\text{EAS}}^* += V_{g,\text{min}} - v_{g,x} \quad (16)$$

where $v_{g,x}$ is the body-frame forward ground speed. This prevents the aircraft being blown backwards.

4. **Bounds:** $V_{\text{EAS}}^* = \text{clamp}(V_{\text{EAS}}^*, V_{\text{EAS},\text{min}}, V_{\text{EAS},\text{max}})$.
5. **Slew-rate controller initialisation:** On first call or after TECS reset: $V_{\text{slew}}^* = \max(V_{\text{EAS},\text{min}}, V_{\text{EAS},\text{meas}})$. If the current state is below $V_{\text{EAS},\text{min}}$ (e.g. minimum changed), force the slew state up to the new minimum.
6. **Slew-rate update:** $V_{\text{out}}^* = \text{SlewRate.update}(V_{\text{EAS}}^*, \Delta t)$ with rate limit `ASPD_SP_SLEW_RATE` (default 1 m/s²).

9 get_manual_airspeed_setpoint()

Converts the throttle stick position $\delta_T \in [-1, 1]$ to an EAS setpoint for manual modes:

- **Manual airspeed enabled** (`FW_POS_STK_CONF` bit 1): piecewise linear interpolation:

$$V^* = \text{interpolateNXY}(\delta_T, \{-1, 0, 1\}, \{V_{\text{min}}, V_{\text{trim}}, V_{\text{max}}\}) \quad (17)$$

Neutral stick ($\delta_T = 0$) gives trim airspeed.

- **Airspeed override** (`DO_CHANGE_SPEED` received): use the commanded value clamped to $[V_{\text{min}}, V_{\text{max}}]$.
- **Default:** return trim airspeed.

10 getManualHeightRateSetpoint()

Converts pitch-stick position to a height rate setpoint for manual altitude control:

$$\dot{h}^* = \begin{cases} \delta_{\text{pitch}} \cdot \dot{h}_{\text{climb,target}} & \delta_{\text{pitch}} > 0 \\ -\delta_{\text{pitch}} \cdot \dot{h}_{\text{sink,target}} & \delta_{\text{pitch}} < 0 \\ \text{NaN} & |\delta_{\text{pitch}}| < \varepsilon_{\text{dead}} \end{cases} \quad (18)$$

NaN output means “no height-rate control” (TECS tracks altitude reference).

11 tecs_update_pitch_throttle()

Single call site for TECS. Called by every position control sub-mode.

Step 1: Inject limits into TECS.

```
1 _tecs.set_min_pitch(pitch_min_rad);
2 _tecs.set_max_pitch(pitch_max_rad);
3 _tecs.set_throttle_min(thr_min);
4 _tecs.set_throttle_max(thr_max);
5 _tecs.set_target_climbrate(target_climbrate);
6 _tecs.set_target_sinkrate(target_sinkrate);
```

Step 2: Low-height mode. If `is_low_height` (aircraft below `FW_LH_APC_FACTOR` × landing approach clearance altitude), TECS uses tighter altitude error gains and smaller pitch limits to improve precision near the ground.

Step 3: Underspeed handling. If `disable_underspeed_handling = false` (default), the `detect_underspeed_enabled` flag is true in TECS.

Step 4: Height-rate override. If the caller provides a finite `height_rate_setpoint`, it is passed to the TECS reference model as a direct height-rate command (bypassing the altitude proportional controller).

Step 5: EAS → TAS setpoint. TECS receives the raw EAS setpoint; internally calls `calcTrueAirspeedSetpoint()` (see Report 2).

Step 6: Call TECS::update().

Step 7: Output.

- `get_tecs_pitch()` returns the TECS pitch setpoint θ^* .
- `get_tecs_thrust()` returns the TECS throttle δ_t^* clamped to $[\delta_{t,\text{min}}, \delta_{t,\text{max}}]$.

12 control_auto(): Setpoint Type Dispatch

control_auto() runs when in FW_POSCTRL_MODE_AUTO.

12.1 handle_setpoint_type()

Before the inner switch, the setpoint type is re-evaluated:

- If velocity control only (no position): return SETPOINT_TYPE_VELOCITY.
- If type is POSITION and not approaching VTOL backtransition: compute 3D distance to waypoint. If aircraft is within the acceptance radius AND has the correct vertical separation:

$$d_{xy} < r_{acc}, \quad d_z > r_{z,acc} \Rightarrow \text{switch to LOITER} \quad (19)$$

r_{acc} comes from `_npfg.switchDistance(500)` which scales with airspeed.

12.2 Inner Switch in control_auto()

```
1 switch (position_sp_type) {
2   case SETPOINT_TYPE_IDLE:      thrust = 0; break;
3   case SETPOINT_TYPE_POSITION:
4     control_auto_position(...); break;
5   case SETPOINT_TYPE_VELOCITY:
6     control_auto_velocity(...); break;
7   case SETPOINT_TYPE_LOITER:
8     control_auto_loiter(...);   break;
9   #ifndef CONFIG_FIGURE_OF_EIGHT
10    case SETPOINT_TYPE_FIGURE_EIGHT:
11      controlAutoFigureEight(...); break;
12  #endif
13 }
```

13 control_auto_position()

1. **Altitude First-Order Hold (FOH):** Linearly interpolates the altitude setpoint between previous and current waypoints as the aircraft approaches:

$$h^* = h_{prev} + \frac{h_{curr} - h_{prev}}{d_{A \rightarrow B} - r_{acc}} \cdot (d_{A \rightarrow B} - d_{min}) \quad (20)$$

where d_{min} is the smallest distance to the current waypoint ever achieved. This ensures the aircraft reaches the waypoint altitude precisely at the acceptance radius rather than demanding a step change.

2. **NPFG guidance:** Calls `navigateWaypoints(prev, curr, ...)` if a valid previous waypoint exists; else `navigateWaypoint(curr, ...)`. Outputs roll setpoint ϕ^* .
3. **Airspeed setpoint:** `adapt_airspeed_setpoint(pos_sp_curr.cruising_speed, ...)`. After NPFG: $V^* \leftarrow \text{getAirspeedRef}() / k$ (NPFG may increase the airspeed demand to maintain track in crosswinds).

4. **TECS call:** `tecs_update_pitch_throttle(h^*, V^*, ...)`.

5. **Attitude setpoint:** $\mathbf{q}^* = \text{Euler}(\phi_{NPFG}^*, \theta_{TECS}^*, \psi)$.

14 control_auto_loiter()

1. **Radius:** from `pos_sp_curr.loiter_radius`; if zero, use `NAV_LOITER_RAD` (default 50m).
2. **Early landing configuration:** If the next setpoint is type LAND and the aircraft is within $R + r_{switch}$ of the loiter circle centre, and `FW_LND_EARLYCFG` enabled:
 - Set airspeed to landing airspeed `FW_LND_AIRSPD` (or stall margin if not set).
 - Deploy flaps: `_flaps_setpoint = FW_FLAPS_LND_SCL`.
 - Deploy spoilers: `_spoilers_setpoint = FW_SPOILERS_LND`.
 - Lower gear: `GEAR_DOWN`.
 - Enable low-height TECS mode.
3. **Landing abort override:** If `_landing_abort_status` is set:
 - If altitude delta to loiter $< kClearanceAltitudeBuffer$ (10m): clear abort status (done).
 - Else: set roll to 0 (fly straight), retract flaps if airspeed is safe.
4. **NPFG:** `navigateLoiter(centre, pos, R, direction, ...)` uses the loiter-circle guidance.
5. **TECS call:** with loiter altitude setpoint.
6. **Attitude setpoint:** $\mathbf{q}^* = \text{Euler}(\phi_{NPFG}^*, \theta_{TECS}^*, \psi)$.

15 control_auto_velocity()

Used for offboard velocity setpoints (no position, only velocity).

1. Target velocity from `pos_sp_curr.vx/vy`; heading from `atan2(vy, vx)`.
2. NPFG navigates toward a virtual waypoint in the target direction.
3. TECS controls altitude to `pos_sp_curr.alt`.

16 control_auto_path()

Used for offboard path following with position + velocity + acceleration.

1. If acceleration is available: compute lateral curvature from the cross-product of normalised velocity and acceleration; use signed loiter radius from centripetal acceleration.
2. NPFG path follows the straight segment defined by velocity direction.
3. TECS controls altitude.

17 control_auto_fixed_bank_alt_hold() — GPS Failsafe Phase 1

Activated when the GPS position estimate is lost (FW_POSCTRL_MODE_AUTO_ALTITUDE) for up to NAV_GPSF_LT seconds (default 30s).

- Roll set to NAV_GPSF_R (fixed open-loop bank angle, default 15°): aircraft circles indefinitely.
- Yaw: 0 (heading holds whatever it was).
- TECS controls altitude to current measured altitude, trim airspeed.
- Special: if NED z or v_z is invalid (no barometer): thrust forced to FW_THR_MIN to avoid climb-away.

18 control_auto_descend() — GPS Failsafe Phase 2

Activated after NAV_GPSF_LT timeout.

- Height rate setpoint hard-coded to -0.5 m/s .
- Same fixed-bank open-loop roll as Phase 1.
- TECS passed this height rate directly (bypasses altitude controller).
- If v_z invalid: thrust forced to minimum.

19 control_auto_takeoff()

Multi-phase launch sequence (catapult, runway, VTOL front-transition):

1. **Pre-launch:** no airspeed, no attitude control; wait for acceleration signature or RC trigger.
2. **Climbout:** TECS active, pitch clamped above FW_TKO_PITCH_MIN to ensure ground clearance. NPFG follows the runway heading (or waypoint).
3. **Transition altitude reached:** gear up; switch to normal auto-position mode.

20 control_auto_landing_straight() and _circular()

20.1 control_auto_landing_straight() — Mission Landing

1. Uses previous WP as the glideslope start; current WP as touchdown.
2. Glideslope: altitude setpoint linearly interpolated from approach altitude to runway elevation.
3. Flare: when within FW_LND_FLALT of touchdown:

- Throttle cut (FW_THR_MIN).
- Pitch held at FW_LND_FL_PMAX.
- Airspeed setpoint switches to FW_LND_AIRSPD.

20.2 control_auto_landing_circular() — Hold Pattern Landing

Used when no previous waypoint is available (e.g. Return-to-Land mode). The aircraft spirals down in a loiter until below the decision altitude, then transitions to straight final.

21 control_manual_altitude()

1. **updateManualTakeoffStatus():** detect first climb above takeoff threshold; set `_completed_manual_takeoff`.
2. Airspeed setpoint: `adapt_airspeed_setpoint(get_manual_airspeed_setpoint(), ...)`.
3. Height rate: `getManualHeightRateSetpoint()` from pitch stick.
4. Pitch minimum: FW_P_LIM_MIN normally; or MIN_PITCH_DURING_MANUAL_TAKEOFF (8°) if not yet completed takeoff.
5. Throttle maximum: if on ground and stick below THROTTLE_THRESH (0.05): $\delta_{t,\max} = 0$ (kill switch).
6. Roll: direct from stick scaled by FW_R_LIM.
7. TECS + attitude setpoint.

22 control_manual_position()

Manual position mode adds heading hold on top of manual altitude:

1. Same airspeed and height-rate processing as altitude mode.
2. **Heading hold engagement condition:** Stick roll and yaw both below HDG_HOLD_MAN_INPUT_THRESH (0.01) AND yaw rate below HDG_HOLD_YAWRATE_THRESH (0.15 rad/s).
3. When engaged:
 - Record hold yaw ψ_{hold} and hold position $\mathbf{p}_{\text{hold}}$.
 - NPFG `navigateLine()` from hold position along heading ψ_{hold} .
 - Roll from NPFG; yaw from ψ_{hold} .
4. EKF XY reset during heading hold: update hold position for 2s to let EKF converge.
5. When stick moves: disable heading hold, use direct roll stick.

23 control_backtransition_heading_hold() and _line_follow()

Used for VTOL back-transition (forward flight → hover).

23.1 heading_hold

- Lock the heading at transition start.
- Simulate wind = 0; ground speed = airspeed aligned with heading.
- NPFG navigates a virtual waypoint 200m ahead.
- Thrust forced to FW_THR_MIN (overridden by transition logic in VTOL module).

23.2 line_follow

- NPFG follows the line from back-transition start position to land waypoint.
- Position at back-transition start captured on first call.

24 Throttle Output (Original)

After the mode dispatch, throttle is applied:

```
1 if (position_sp_type == SETPOINT_TYPE_IDLE) {
2     _att_sp.thrust_body[0] = 0.0f;
3 } else if (_landed && ...) {
4     _att_sp.thrust_body[0] = min(_param_fw_thr_idle.get(), 1.f);
5 } else {
6     _att_sp.thrust_body[0] = min(get_tecs_thrust(), _param_fw_thr_max.get());
7 }
```

No ramp logic. TECS output goes directly to the airframe actuators.

Part II

Soaring-Mode Modifications and Their Physical Impact

25 Overview of Modified Functions

The soaring implementation modifies the following functions:

Table 1: Summary of function-level changes

Function	What changes
Run()	DDS poll, inhibit/watchdog, status republish, DDS→cmd arbiter
vehicle_command_poll()	CUSTOM_0, DO_SET_MODE, publish_autosoaring_status() on soar:off
publish_autosoaring_status()	uORB autosoaring_status bridged to ROS 2
parameters_update()	Glide airspeed eager pre-push
control_auto()	Full soaring dispatch block
Throttle output	Symmetric entry/exit ramps

26 Change 1: Run() — DDS Poll, Inhibit, and Stale-ness Watchdog

26.1 DDS Poll

```
1 autosoaring_control_s soaring_msg;
2 if (_autosoaring_control_sub.update(&soaring_msg)) {
3     const hrt_abstime now = hrt_absolute_time();
4     const bool dds_inhibited = (_soaring_dds_inhibit_until_us > 0
5     && now < _soaring_dds_inhibit_until_us);
6     const bool dds_wants_soaring =
7         (soaring_msg.soaring_mode != autosoaring_control_s::SOARING_OFF)
8         ;
9
10    if (dds_inhibited && dds_wants_soaring) {
11        // ignore: CLI soar:off inhibit active
12    } else {
13        if (!dds_wants_soaring) {
14            _soaring_dds_inhibit_until_us = 0; // DDS OFF cancels
15            inhibit
16        }
17        if (!_soaring_local_override) {
18            const bool prev_thermal = (
19                _autosoaring_control.soaring_mode ==
20                autosoaring_control_s::SOARING_THERMAL_LOITER ||
21                _autosoaring_control.soaring_mode ==
22                autosoaring_control_s::SOARING_THERMAL_BANK);
23            const bool new_thermal = (
24                soaring_msg.soaring_mode == autosoaring_control_s::
25                SOARING_THERMAL_LOITER ||
26                soaring_msg.soaring_mode == autosoaring_control_s::
27                SOARING_THERMAL_BANK);
28            _autosoaring_control = soaring_msg;
29            _autosoaring_last_rcv_us = now;
30            if (!prev_thermal && new_thermal) {
31                _soaring_mode_cmd_last_us = 0;
32                _soaring_last_lat = NAN;
33                _soaring_last_lon = NAN;
34            }
35        }
36    }
37 }
```

31

}

DDS inhibit: When `soar:off` is issued from CLI, the DDS path is blocked for 5s (`_soaring_dds_inhibit_until_us = now + 5s`). This prevents an autonomous companion from immediately re-enabling soaring after the operator has manually stopped it. If the companion publishes `soaring_mode = OFF` while the inhibit is active, the remaining cooldown is cleared immediately (`_soaring_dds_inhibit_until_us = 0`).

Authority arbitration: The `_soaring_local_override` flag gives CLI exclusive authority. When set, DDS messages are received but do *not* overwrite `_autosoaring_control`. CLI and DDS cooperate: CLI sets the mode; DDS can update ancillary fields (loiter position, bank angle) while CLI holds authority.

26.2 Staleness Watchdog

```

1 if (!_soaring_local_override &&
2   _autosoaring_last_recv_us > 0 &&
3   (hrt_absolute_time() - _autosoaring_last_recv_us) > 2_s) {
4   if (_autosoaring_control.soaring_mode != autosoaring_control_s::
5     SOARING_OFF) {
6     _autosoaring_control.soaring_mode = autosoaring_control_s::
7       SOARING_OFF;
8     publish_autosoaring_status(autosoaring_status_s::
9       SOURCE_STALENESS_WATCHDOG,
10      autosoaring_control_s::SOARING_OFF, 0);
11   }
12   _tecs.set_gliding_mode_enabled(false);
13   _autosoaring_last_recv_us = 0;
14 }

```

Physical motivation: The companion computer may crash or lose its DDS connection while in mid-glide. Without a watchdog, the aircraft would remain engine-off indefinitely at risk of altitude loss. The 2s timeout restores powered flight automatically.

Not triggered by CLI: `_autosoaring_last_recv_us = 0` when CLI is the authority (set in `vehicle_command_poll()` for `CUSTOM_0` commands). The watchdog only fires if `_autosoaring_last_recv_us > 0`, which is only set when DDS delivered a message.

26.3 AutosoaringStatus publication, 1 Hz inhibit republish, and 2 Hz phase heartbeat

On CLI `soar:off` (`CUSTOM_0` with mode 0), after TECS gliding is disabled, `publish_autosoaring_status()` is called with `SOURCE_CLI_SOARING_OFF`, the effective `soaring_mode` (OFF), `dds_inhibit_until_us` equal to `hrt_absolute_time() + 5s`, and `fm_u_soaring_phase = SOARING_PHASE_CRUISE`. The timestamp `_autosoaring_status_repub_us` is initialised so that, for the duration of the inhibit window, the same status is republished at 1 Hz (companion may miss a single DDS sample):

```

1 if (_soaring_dds_inhibit_until_us > 0 && status_now <
2   _soaring_dds_inhibit_until_us) {
3   if (status_now - _autosoaring_status_repub_us >= 1_s) {
4     _autosoaring_status_repub_us = status_now;
5   }
6 }

```

```

4   publish_autosoaring_status(SOURCE_CLI_SOARING_OFF,
5     _autosoaring_control.soaring_mode,
6     _soaring_dds_inhibit_until_us,
7     SOARING_PHASE_CRUISE);
8 }
9 } else {
10   _autosoaring_status_repub_us = 0;
11 }

```

In addition, a 2 Hz periodic heartbeat is published at the end of every `control_auto()` call (inside the position-control rate loop at ≈ 50 Hz), derived from the current `effective_glide/effective_thermal` booleans:

```

1 if (phase_now - _autosoaring_status_periodic_us >= 500_ms) {
2   _autosoaring_status_periodic_us = phase_now;
3   uint8_t fmu_phase;
4   if (effective_thermal)
5     fmu_phase = PX4_ISFINITE(_soaring_last_lat)
6       ? SOARING_PHASE_THERMAL : SOARING_PHASE_SEARCHING;
7   else if (effective_glide)
8     fmu_phase = SOARING_PHASE_GLIDING;
9   else
10    fmu_phase = SOARING_PHASE_CRUISE;
11   publish_autosoaring_status(SOURCE_PERIODIC,
12     _autosoaring_control.soaring_mode,
13     _soaring_dds_inhibit_until_us, fmu_phase);
14 }

```

This means `AutosoaringStatus` is **not** limited to the 5s inhibit window. It is published continuously: every 500ms during any active soaring, at 1Hz during the DDS inhibit cooldown, and as an immediate one-shot on event (watchdog or `soar:off`). Because uORB automatically logs every published topic, the phase sequence is available in the ULog for post-flight analysis.

The uORB topic is listed under publications in `dds_topics.yaml` as `/fmu/out/autosoaring_status`.

27 Change 2: vehicle_command_poll() — CUSTOM_0 and DO_SET_MODE

27.1 VEHICLE_CMD_CUSTOM_0 (Soaring CLI)

```

1 const uint8_t new_mode = constrain((int)cmd.param1,
2   0, (int)SOARING_THERMAL_BANK);
3 _autosoaring_control.soaring_mode = new_mode;
4 _autosoaring_control.loiter_radius_m =
5   (cmd.param3 > FLT_EPSILON) ? cmd.param3 : NAN;
6 if (PX4_ISFINITE(cmd.param2) && cmd.param2 > FLT_EPSILON)
7   _autosoaring_control.bank_angle_cmd = cmd.param2;

```

On enabling (`new_mode != SOARING_OFF`):

- `_soaring_local_override = true`
- Clear inhibit (`_soaring_dds_inhibit_until_us = 0`)

- Clear latch (`_soaring_forbidden_latched = false`)
- Disable watchdog (`_autosoaring_last_rcv_us = 0`)
- Set “paired mode-switch” flag (`_soaring_expect_set_mode = true`)
- Reset DO_REPOSITION debounce (`_soaring_mode_cmd_last_us = 0, _soaring_last_lat = NAN`)

On disabling (`new_mode == SOARING_OFF`):

- Clear override and latch
- Block DDS for 5s (`_soaring_dds_inhibit_until_us = now + 5s`)
- Immediately call `_tecs.set_gliding_mode_enabled(false)`
- Publish `autosoaring_status` (`SOURCE_CLI_SOARING_OFF`, inhibit deadline) and seed the 1 Hz republish timer

27.2 VEHICLE_CMD_DO_SET_MODE

When CLI has override, the module must distinguish between:

1. A `DO_SET_MODE` *paired* with `CUSTOM_0` (i.e. `soar:glide` sends both in sequence): the mode switch IS part of soaring activation. Consume `_soaring_expect_set_mode` and keep soaring on.
2. A standalone `DO_SET_MODE` (operator typed commander mode `auto:mission`): interpret as “exit soaring, restore powered flight”.

```

1 if (_soaring_local_override) {
2     if (_soaring_expect_set_mode) {
3         _soaring_expect_set_mode = false; // consume pairing
4     } else {
5         // Check target mode: ignore QGC heartbeats (AUTO_MISSION/LOITER)
6
7         const uint8_t main = (uint8_t)cmd.param2;
8         const uint8_t sub = (uint8_t)cmd.param3;
9         const bool compat =
10             (main == PX4_CUSTOM_MAIN_MODE_AUTO &&
11              (sub == PX4_CUSTOM_SUB_MODE_AUTO_MISSION ||
12               sub == PX4_CUSTOM_SUB_MODE_AUTO_LOITER));
13         if (!compat) {
14             // Incompatible mode (MANUAL, STABILIZED, etc.) -> exit
15             // soaring
16             _autosoaring_control.soaring_mode = SOARING_OFF;
17             _soaring_local_override = false;
18             _tecs.set_gliding_mode_enabled(false);
19         }
20         // else: AUTO_MISSION / AUTO_LOITER heartbeat -> keep soaring
21     }
22 }
```

QGC heartbeat fix. QGC periodically broadcasts `DO_SET_MODE(AUTO_MISSION)` to maintain the displayed mode. Without the `compat` guard, every such heartbeat would arrive as a “standalone” mode-switch and immediately disable soaring — a race condition that was observed in SITL testing.

The guard keeps soaring active when the target mode is `AUTO_MISSION` or `AUTO_LOITER` (both are soaring-compatible). Any other mode (`MANUAL`, `ACRO`, `STABILIZED`, `ALTITUDE`, `POSCtl`) is considered an intentional operator override and correctly exits soaring.

Interaction with altitude floor. The QGC fix keeps `_soaring_local_override = true` permanently during CLI glide, which had the unintended side effect of also keeping the `FW_ALT_MIN` check bypassed (the old code guarded it with `!_soaring_local_override`). This was corrected by removing that guard from the altitude floor check (see Step 2 above).

28 Change 3: parameters_update() — Eager Glide Airspeed Push

```

1 const float polar_a = _param_fw_polar_a.get();
2 const float polar_b = _param_fw_polar_b.get();
3 const float v_polar = (polar_a > FLT_EPSILON && polar_b > FLT_EPSILON)
4     ? sqrtf(polar_b / polar_a) : 0.0f;
5 const bool polar_mode =
6     (_autosoaring_control.soaring_mode == SOARING_GLIDE_POLAR);
7 _tecs.set_gliding_airspeed_setpoint(
8     polar_mode && v_polar > FLT_EPSILON
9     ? v_polar : _param_fw_glide_airspd.get());
10 _tecs.set_glide_i_decay(_param_fw_glide_i_decay.get());
11 _tecs.set_glide_stall_eas(_param_fw_airspd_stall.get());
```

Why eager: When Mode 2 (polar) activates mid-mission, there is one TECS cycle between the DDS message arriving and `control_auto()` running. The eager push here ensures that first cycle uses the correct polar setpoint rather than the old fixed value.

FW_AIRSPD_STALL push: `set_glide_stall_eas()` stores the stall speed so that `calcTrueAirspeedSetpoint()` can use it as the glide airspeed floor instead of a hard-coded 3m/s magic constant. `FW_AIRSPD_STALL` already exists in PX4 and is used by `fw_att_control` and `airspeed_selector`. `fw_pos_control` now reads it and forwards it to TECS on every parameter update.

29 Change 4: control_auto() — Full Soaring Dispatch Block

This block executes at the top of `control_auto()` before the inner setpoint-type switch.

29.1 Step 1: Extract Mode Variables

```

1 const uint8_t soaring_cmd = _autosoaring_control.soaring_mode;
2 const bool glide_cmd = (soaring_cmd == SOARING_GLIDE_FIXED ||
3     soaring_cmd == SOARING_GLIDE_POLAR);
```

```

4 const bool thermal_cmd    = (soaring_cmd == SOARING_THERMAL_LOITER ||
5   soaring_cmd == SOARING_THERMAL_BANK);
6 const bool soaring_allowed = soaring_cmd != SOARING_OFF &&
7   (_soaring_local_override || !_soaring_forbidden_latched);

```

29.2 Step 2: Altitude Floor (FW_ALT_MIN)

```

1 // Enforced on ALL paths (DDS and CLI).
2 if (soaring_requested && current_altitude < alt_min) {
3     if (!_soaring_forbidden_latched) {
4         _soaring_forbidden_latched = true;
5         if (_soaring_local_override) {
6             // CLI was in control: release override so mission resumes
7             _soaring_local_override = false;
8             _soaring_expect_set_mode = false;
9             PX4_WARN("Soaring_CLI_override_released: alt %.0f < FW_ALT_MIN %.0f",
10               current_altitude, alt_min);
11         } else {
12             PX4_WARN("Soaring_disabled: alt %.0f < min %.0f",
13               current_altitude, alt_min);
14         }
15     }
16 }
17 if (!soaring_requested)
18     _soaring_forbidden_latched = false;

```

Design rationale. The latch prevents rapid cycling near the floor altitude: once `_soaring_forbidden_latched` is set it stays set until soaring is explicitly disabled (`soaring_requested = false`).

CLI mode fix (hardware testing). The original code had:

```

1 if (!_soaring_local_override && soaring_requested && ...)

```

This skipped the check when CLI was active (`_soaring_local_override = true`). The intent was “the pilot is in control, trust them”. However, a side-effect of the QGC-heartbeat fix (which keeps `_soaring_local_override` permanently true during CLI glide) was that the altitude floor was *never* triggered on hardware. The aircraft descended to ground without automatic recovery.

Current design: the check fires for *all* paths. When it fires during CLI override, `_soaring_local_override` is also cleared so that:

1. TECS restores throttle immediately (next cycle: `effective_glide = false`).
2. The glide-exit block (Step 9) fires and publishes `DO_SET_MODE → AUTO_MISSION`.
3. The operator must re-issue `soar:glide` after regaining altitude.

Altitude measurement.

```

1 const float current_altitude =
2   _local_pos.z_global
3   ? -_local_pos.z + _local_pos.ref_alt
4   : _current_altitude;

```

When global-frame position is available (`z_global = true`), the GPS altitude is used. Otherwise the barometric estimate `_current_altitude` is used. Both are above mean sea level (AMSL) — set `FW_ALT_MIN` accordingly (e.g. home AMSL + minimum AGL).

29.3 Step 3: Altitude Ceiling with Hysteresis

$$h \geq h_{\max} \Rightarrow \text{_alt_max_reached} = \text{true} \quad (21)$$

$$h < h_{\max} - h_{\text{hyst}} \Rightarrow \text{_alt_max_reached} = \text{false} \quad (22)$$

Where $h_{\max} = \text{FW_ALT_MAX}$, $h_{\text{hyst}} = \text{FW_ALT_HYST}$.

When ceiling is reached during thermalling, mode degrades to glide:

$$\text{effective_thermal} = \text{false}, \quad \text{effective_glide} = \text{true} \quad (23)$$

29.4 Step 4: Effective Mode Flags

```

1 const bool effective_glide =
2   soaring_allowed && (glide_cmd || _alt_max_reached);
3 const bool effective_thermal =
4   soaring_allowed && thermal_cmd && !_alt_max_reached;
5 const bool effective_thermal_bank =
6   effective_thermal && (soaring_cmd == SOARING_THERMAL_BANK);
7 const bool currently_soaring = effective_glide || effective_thermal;

```

29.5 Step 5: TECS Flag and Airspeed Injection

```

1 _tecs.set_gliding_mode_enabled(currently_soaring);
2
3 if (soaring_cmd == SOARING_GLIDE_FIXED
4   && _autosoaring_control.airspeed_cmd > FLT_EPSILON)
5     _tecs.set_gliding_airspeed_setpoint(
6       _autosoaring_control.airspeed_cmd); // Mode 1
7 else if (soaring_cmd == SOARING_GLIDE_POLAR) {
8     const float v = sqrtf(_param_fw_polar_b.get() /
9       _param_fw_polar_a.get());
10    _tecs.set_gliding_airspeed_setpoint(v); // Mode 2
11 } else
12    _tecs.set_gliding_airspeed_setpoint(
13      _param_fw_glide_airspd.get()); // Modes 3,4
14
15 // Bank angle for load-factor stall floor
16 _soaring_bank_angle_cmd_deg =
17   (PX4_ISFINITE(_autosoaring_control.bank_angle_cmd) &&
18     _autosoaring_control.bank_angle_cmd > FLT_EPSILON)
19   ? _autosoaring_control.bank_angle_cmd
20   : _param_fw_thermal_bank.get();
21 _tecs.set_load_factor(1.f / cosf(
22   radians(_soaring_bank_angle_cmd_deg)));

```

29.6 Step 6: Thermal DO_REPOSITION Command

When `effective_thermal` is true, the navigator must switch to `AUTO_LOITER`.

29.6.1 Loiter Centre Pinning

```

1 const bool dds_has_pos =
2     PX4_ISFINITE(_autosoaring_control.loiter_lat) &&
3     fabsf(_autosoaring_control.loiter_lat) > FLT_EPSILON;
4 const double lat = dds_has_pos
5     ? (double)_autosoaring_control.loiter_lat : curr_pos(0);
6 const double lon = dds_has_pos
7     ? (double)_autosoaring_control.loiter_lon : curr_pos(1);
8 const bool first_activation = !PX4_ISFINITE(_soaring_last_lat);
9 const bool pos_changed = first_activation || explicit_pos_changed;
10 const bool dir_changed = !first_activation &&
11     (_autosoaring_control.loiter_clockwise !=
12     _soaring_last_clockwise);

```

Bug fix: Without pinning, `DO_REPOSITION` is sent every cycle with the drifting `curr_pos`, causing a moving loiter centre. The `first_activation` flag sends exactly *one* `DO_REPOSITION`. Subsequent cycles: `pos_changed = false`, command not re-sent.

29.6.2 T3: Pre-Roll Airspeed Safety

On the first activation only:

$$n_z = \frac{1}{\cos|\phi_{\text{cmd}}|}, \quad V_{\text{floor}} = 1.05 \times V_{\text{stall,level}} \times \sqrt{n_z} \quad (24)$$

If $V_{\text{EAS,meas}} < V_{\text{floor}}$: delay the `DO_REPOSITION`. The aircraft stays in glide until speed is sufficient.

29.6.3 Mission Index Latch

Before the first `DO_REPOSITION`:

```

1 mission_s snap{};
2 if (_mission_sub.copy(&snap) && snap.count > 0) {
3     _soaring_mission_resume_seq = (uint16_t)constrain(
4         snap.current_seq, 0, snap.count - 1);
5     _soaring_mission_resume_valid = true;
6 }

```

29.6.4 Minimum Radius and Signed Encoding

$$R_{\text{min}} = \frac{(V_{\text{EAS,glide}} \cdot k)^2}{g \tan|\phi_{\text{cmd}}|} \quad (25)$$

$$R_{\text{eff}} = \max(\text{loiter_radius_m}, R_{\text{min}}) \quad (26)$$

$$\text{cmd.param3} = \begin{cases} +R_{\text{eff}} & \text{loiter_clockwise} = \text{true} \\ -R_{\text{eff}} & \text{loiter_clockwise} = \text{false} \end{cases} \quad (27)$$

29.6.5 Complete DO_REPOSITION Fields

Field	Value
command	VEHICLE_CMD_DO_REPOSITION
param1	-1 (keep current speed)
param2	1 (switch to loiter)
param3	$\pm R_{\text{eff}}$ (signed radius)
param4	NaN (keep yaw)
param5	loiter lat [deg]
param6	loiter lon [deg]
param7	current altitude [m AMSL]

29.7 Step 7: Glide DO_SET_MODE Heartbeat

```

1 if (effective_glide &&
2     (now - _soaring_mode_cmd_last_us) > 1_s) {
3     cmd.param2 = (float)PX4_CUSTOM_MAIN_MODE_AUTO;
4     cmd.param3 = (float)PX4_CUSTOM_SUB_MODE_AUTO_MISSION;
5     _pub_vehicle_command.publish(cmd);
6     _soaring_mode_cmd_last_us = now;
7 }

```

Sent every second to keep navigator in `AUTO_MISSION` while gliding.

29.8 Step 8: Thermal Exit

```

1 if (_soaring_was_thermal && !effective_thermal && !effective_glide) {
2     // switch back to mission
3     exit_cmd.param3 = (float)PX4_CUSTOM_SUB_MODE_AUTO_MISSION;
4     _pub_vehicle_command.publish(exit_cmd);
5     // clear all state
6     _soaring_last_lat = NAN;
7     _soaring_last_lon = NAN;
8     _soaring_last_clockwise = true;
9     _soaring_local_override = false;
10    // schedule mission resume next cycle
11    if (_soaring_mission_resume_valid)
12        _soaring_pending_mission_restore = true;
13 }

```

Mission restore is applied one cycle later, after the navigator has confirmed it is in `AUTO_MISSION` state:

```

1 if (_soaring_pending_mission_restore
2     && nav_state == NAVIGATION_STATE_AUTO_MISSION) {
3     restore_cmd.command = VEHICLE_CMD_MISSION_START;
4     restore_cmd.param1 = (float)_soaring_mission_resume_seq;
5     _pub_vehicle_command.publish(restore_cmd);
6     _soaring_pending_mission_restore = false;
7 }

```

29.9 Step 9: Glide Exit — DO_SET_MODE → AUTO_MISSION

This block is new and mirrors Step 8 for the *glide* case. When the aircraft was gliding (`_soaring_was_glide = true`) and glide just ended (altitude floor, explicit `soar:off`, or any other path), the FMU publishes a mode-switch command so the navigator re-enters mission flight:

```
1 const bool soaring_was_glide_only =
2     _soaring_was_glide && !_soaring_was_thermal;
3
4 if (soaring_was_glide_only && !effective_glide && !effective_thermal) {
5     vehicle_command_s cmd{};
6     cmd.command = VEHICLE_CMD_DO_SET_MODE;
7     cmd.param1 = 1.f; // CUSTOM_MODE_ENABLED
8     cmd.param2 = (float)PX4_CUSTOM_MAIN_MODE_AUTO;
9     cmd.param3 = (float)PX4_CUSTOM_SUB_MODE_AUTO_MISSION;
10    _pub_vehicle_command.publish(cmd);
11
12    // Clear override so vehicle_command_poll() does not intercept
13    // this DO_SET_MODE as a standalone "exit soaring" command.
14    _soaring_local_override = false;
15    _soaring_forbidden_latched = false;
16    PX4_INFO("Autosoaring: glide ended-> AUTO_MISSION");
17 }
```

Why is this needed? In *thermal* mode the aircraft is in AUTO_LOITER; the thermal-exit block (Step 8) must explicitly switch back to AUTO_MISSION. In *glide* mode the aircraft stays in AUTO_MISSION throughout, but the navigator has been receiving continuous DO_SET_MODE(AUTO_MISSION) heartbeats from the 1s refresh. When glide ends the navigator is already in AUTO_MISSION — yet republishing the command ensures the navigator re-evaluates its setpoints and resumes waypoint tracking immediately, rather than coasting on the last glide setpoint.

Guard: `soaring_was_glide_only`. The flag `_soaring_was_glide` is updated at the end of each `control_auto()` cycle:

```
1 _soaring_was_glide = effective_glide;
2 _soaring_was_thermal = effective_thermal;
3 _soaring_was_bank_thermal = effective_thermal_bank;
```

The `!_soaring_was_thermal` guard prevents a double fire when a thermal-loiter ends and simultaneously `effective_glide` is false: the thermal-exit block (Step 8) already handles that case.

Interaction with `vehicle_command_poll()`. Clearing `_soaring_local_override` before publishing ensures that when the published DO_SET_MODE loops back through `vehicle_command_poll()`, it is processed as a normal mode switch and does not trigger the “standalone exit soaring” branch (which would call `set_gliding_mode_enabled(false)` redundantly).

30 Change 5: Mode 4 — Complete NPFG Bypass

30.1 Why Bypass

If the inner switch runs `control_auto_loiter()` first:

- NPFG computes `getAirspeedRef()` based on loiter geometry and wind.
- This modified airspeed propagates through `adapt_airspeed_setpoint()` and into TECS.
- TECS pitch then tracks the geometry-contaminated airspeed, not the polar or fixed glide airspeed.
- Result: airspeed oscillations and pitch-roll coupling.

30.2 Implementation

```
1 if (effective_thermal_bank) {
2     // Skip the switch entirely
3     tecs_update_pitch_throttle(
4         control_interval, current_altitude,
5         _param_fw_glide_airspeed.get(), // pure glide speed
6         radians(_param_fw_p_lim_min.get()),
7         radians(_param_fw_p_lim_max.get()),
8         _param_fw_thr_min.get(), _param_fw_thr_max.get(),
9         _param_sinkrate_target.get(), _param_climbrate_target.get(),
10        false);
11
12    const float bank_sign =
13        _autosoaring_control.loiter_clockwise ? 1.f : -1.f;
14    const float bank_rad =
15        bank_sign * radians(fabsf(_soaring_bank_angle_cmd_deg));
16    const Quatf q(Eulerf(bank_rad, get_tecs_pitch(), _yaw));
17    q.copyTo(_att_sp.q_d);
18    _att_sp.thrust_body[0] = get_tecs_thrust(); // = 0 (glide)
19
20 } else {
21     switch (position_sp_type) { /* standard dispatch */ }
22 }
```

The quaternion **q*** encodes exactly three independently sourced values:

Component	Source
ϕ^*	$\pm \phi_{\text{cmd}} $ (companion bank angle)
θ^*	<code>get_tecs_pitch()</code> (glide speed hold)
ψ	current measured yaw (inner loop handles yaw coordination)

31 Change 6: Symmetric Throttle Ramps

31.1 Physical Motivation

Entry (powered → glide), instant cut:

- Propwash instantly collapses, reducing wing root dynamic pressure.
- Nose-down pitching moment, then TECS corrects: ± 1.5 m/s airspeed transient.

Exit (glide $\rightarrow$ powered), instant restore:

- Propwash surge + pitch integrator bias: 40–50 deg/s spike.

Both transients resolved by linear ramps of duration $T = \text{FW_GLIDE_RAMP_T}$ (default 3 s).

31.2 Pre-Glide Thrust Cache

Before glide starts, the current TECS thrust is saved:

```
1 if (!currently_soaring && !_landed)
2   _soaring_pre_glide_thrust = get_tecs_thrust();
```

31.3 Entry Ramp (glide starts)

On glide entry, `_soaring_entry_ramp_start_us` is set. Each cycle during entry:

$$\delta_t^{\text{entry}}(t) = \delta_{t,\text{pre}} \cdot \max\left(1 - \frac{t - t_s}{T}, 0\right) \quad (28)$$

After T seconds: $\delta_t^{\text{entry}} = 0$.

31.4 Exit Ramp (glide ends)

On glide exit, `_soaring_ramp_start_us` is set. Scale factor:

$$s_{\text{exit}}(t) = \min\left(\frac{t - t_e}{T}, 1\right) \quad (29)$$

31.5 Final Throttle Selection

Condition	Thrust formula	Ramp active?
Idle setpoint	0	No
Currently soaring, entry ramp active	$\delta_{t,\text{pre}}(1 - t/T)$	Entry
Currently soaring, ramp done	0	None (steady glide)
Not soaring, exit ramp active	$\delta_{t,\text{TECS}} \cdot s_{\text{exit}}$	Exit
Not soaring, normal powered	$\delta_{t,\text{TECS}}$	None
Landed	$\min(\delta_{t,\text{idle}}, 1)$	No

31.6 Physical Impact

Metric	Before ramp	After ramp
Entry airspeed deviation	~ 1.5 m/s	$\lesssim 0.4$ m/s
Exit pitch-rate spike	40–50 deg/s	$\lesssim 15$ deg/s
Throttle gradient	Step	$\leq 0.33 \text{ s}^{-1}$ (over $T = 3$ s)

32 Complete Modification Summary

#	Function / Feature	Original	Modified
1	<code>Run()</code> DDS poll	Not present	DDS poll + inhibit/watchdog
2	DDS authority	N/A	CLI overrides DDS; 5 s inhibit on <code>soar:off</code>
3	Staleness watchdog	N/A	2 s silence $\Rightarrow$ SOARING_OFF
4	<code>vehicle_command</code> pinhandled CUSTOM_0		Encodes modes 0–4
5	DO_SET_MODE Ignored handling		Pairing detection + exit logic
6	<code>parameters_update</code> push FW params		Also push glide/polar airspeed
7	<code>control_auto()</code>	Mode switch only	Full soaring dispatch at top
8	Altitude floor	None	Latch below FW_ALT_MIN
9	Altitude ceiling	None	Hysteresis at FW_ALT_MAX
10	Nav (glide)	Not commanded	DO_SET_MODE heartbeat 1 s
11	Nav (thermal)	Not commanded	DO_REPOSITION once (pinned)
12	Loiter direction	N/A	Signed radius, resend on flip
13	Pre-roll safety	None	5% above banked stall
14	Mission index	Not saved	Latched; restored on exit
15	TECS glide flag	Never set	Set for modes 1–4
16	Airspeed setpoint	Mission speed	Mode-dependent (fixed/polar/hold)
17	Mode 4 roll	None	Direct $\pm \phi_{\text{cmd}} $, NPFG bypassed
18	Throttle entry	Instant cut	Linear ramp T_{ramp}
19	Throttle exit	Instant restore	Linear ramp T_{ramp}